Paragliding Intelligence Platform

Volume 12 - Legal, Safety & Data Licensing v2 Technical

Compliance architecture for a paragliding-first meteo, AI, live-tracking and decision-support platform

Document status: Architecture and implementation design; not legal advice.

Prepared for: staged technical specification suite - Volume 12.

1. Purpose and Legal Positioning

This volume defines the legal, safety, licensing and governance architecture for the Paragliding Intelligence Platform. The platform combines weather data, terrain data, airspace data, AI-generated flight briefings, live tracking, paid subscriptions and rescue-oriented features. Because these features influence safety decisions in aviation-like outdoor activity, the product must be designed as a decision-support system with explicit uncertainty, traceability, source attribution and liability boundaries.

The document is written for founders, product owners, engineering leads, security leads, data protection officers, legal counsel and operations staff. It converts legal and safety concerns into system requirements, database fields, API behavior, UI copy, evidence retention and release gates.

This document is not legal advice. It is an engineering compliance blueprint. A qualified lawyer should review local consumer law, aviation law, data-protection obligations, liability wording, subscription terms and partner agreements before commercial launch.

1.1 Product legal thesis

- The platform must not be marketed as an official aviation authority, air traffic control system, certified flight planning system or guaranteed safety service.
- The product should be marketed as weather intelligence, flight-preparation support, risk visualization, community observation and emergency-assistance tooling.
- GO / MAYBE / NO-GO labels are internal recommendation labels, not legal permission to fly.
- Every AI output must be grounded in retrieved weather, site, pilot-profile and rule data; it must not invent missing conditions.
- For missing, stale, conflicting or low-confidence data, the system must degrade to MAYBE or NO-DATA rather than optimistic GO.
- Live tracking, rescue links and emergency contacts require explicit consent and strong privacy controls.

1.2 Compliance-by-design principles

Principle	Engineering implication	Acceptance test
Traceability	Every forecast, score and AI briefing stores source model, model run time, derived variables and policy version.	Given a briefing ID, support can reconstruct the exact source data and rules used.
Non-endorsement	Attribution text must not imply ECMWF, MET Norway, Open-Meteo, OpenAIP, OSM or Copernicus endorse the app.	Attribution component passes source-specific wording review.
Conservative safety	Hard blockers override commercial features and paid-tier entitlements.	Thunderstorm, no-data, stale data and high model-disagreement tests cannot return GO.
Privacy minimization	Default settings collect minimum necessary location and pilot-profile data.	New account can use weather map without live tracking enabled.
Human responsibility	UI copy states that pilot-in-command remains responsible for launch and flight decisions.	All launch-decision screens show concise safety boundary copy.
Revocability	Users can stop live tracking, delete location history, export data and revoke rescue links.	Privacy test suite verifies all controls.

2. Regulatory and Risk Classification

The platform is not a certified avionics product, but it affects flight-safety decisions. The safest commercial posture is to implement controls similar to safety-critical decision support even where not legally required. This creates trust and reduces avoidable liability.

2.1 Regulatory areas

Area	Why it matters	Required architecture response
Data licensing	Weather, terrain, map and airspace datasets have attribution, rate-limit and commercial-use obligations.	Maintain a source registry, attribution renderer, licence review queue and source-specific usage controls.
GDPR / privacy	Live location, track history and emergency contacts are personal data.	Consent, data minimization, retention rules, export/delete workflow, regional data controls and processor contracts.
EU AI Act posture	AI is used for decision support and safety-	Risk classification memo, human oversight,

	related recommendations; the legal classification should be reviewed as the product evolves.	technical documentation, logging, model monitoring and clear AI transparency.
Consumer/subscription law	Paid tiers, renewal, cancellation and refunds are regulated by app stores, Stripe and local consumer rules.	Clear checkout terms, entitlement audit logs, cancellation path and invoice records.
Product liability / negligence	Users may rely on safety recommendations.	No guarantee language, uncertainty display, hard blockers, source freshness, incident audit logs and conservative fallback.
Aviation/airspace rules	Airspace data can be incomplete/stale; local law controls flying.	Airspace disclaimer, source timestamp, no official navigation guarantee and links to authoritative data.
Emergency/rescue workflows	Rescue features may create reliance by users and responders.	Explicit emergency disclaimer, best-effort availability, immutable event logs and clear escalation limitations.

2.2 AI risk-classification memo

Before launch, legal counsel should create and maintain an AI classification memo. The product should be treated internally as safety-relevant decision support even if it is not finally classified as a high-risk AI system. The engineering design must preserve evidence that the system provides information and recommendations, not autonomous control of a regulated aircraft.

AI subsystem	Potential risk	Required control
AI flight briefing	User may rely on narrative summary and ignore raw weather.	Briefing must cite source fields, show confidence, expose blockers and prohibit invented values.
AI XC planner	Route suggestions could lead pilots into unsafe terrain/airspace.	Route output must include uncertainty, airspace checks, landing options and safety downgrade logic.
Digital twin correction	Model may overfit or understate risk at a site.	Model promotion gates, false-green metric, rollback and human review for safety regressions.
Pilot personalization	Beginner/advanced profile changes recommendation.	Transparent profile factors, user correction controls and conservative defaults.
Webcam CV	Computer vision may misread clouds or visibility.	CV signal is secondary; never override hard weather blockers alone.

3. Data Source and Licence Register

The platform must maintain a machine-readable register for every external dataset, API, model, tile provider and licensed component. The register controls ingestion, attribution, redistribution, caching, downstream products and commercial-tier eligibility.

3.1 Canonical source register schema

```
CREATE TABLE data_source_licences (
  id UUID PRIMARY KEY,
  source_key TEXT UNIQUE NOT NULL,
  source_name TEXT NOT NULL,
  source_type TEXT NOT NULL, -- weather_api, raw_model, terrain, map, airspace, station, satellite
  licence_name TEXT,
  licence_url TEXT,
  commercial_use_allowed BOOLEAN,
  redistribution_allowed BOOLEAN,
  attribution_required BOOLEAN NOT NULL DEFAULT TRUE,
  attribution_text TEXT NOT NULL,
  non_endorsement_required BOOLEAN NOT NULL DEFAULT TRUE,
  rate_limit_policy JSONB NOT NULL DEFAULT '{}',
  retention_policy JSONB NOT NULL DEFAULT '{}',
  review_status TEXT NOT NULL DEFAULT 'pending', -- pending, approved, restricted, prohibited
```

```
  reviewed_by TEXT,
  reviewed_at TIMESTAMPTZ,
  notes TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

3.2 Licence matrix

Source	Product use	Commercial posture	Critical obligations
ECMWF Open Data	Raw global forecast model data, ensemble/model comparison, derived forecast cubes.	Usable commercially under CC BY 4.0 for the open-data subset; verify exact product/service tier before ingesting.	Attribution, licence link, indicate modifications, no implied endorsement, respect access limits and service terms.
NOAA/NCEP GFS via NOMADS	Free global model fallback and comparison layer.	Generally open US government data, but operational use still requires attribution, availability fallback and no warranty assumption.	Track model run, product ID, forecast step and NOAA/NCEP source.
MET Norway API	Nordic coordinate forecast, nowcast, warnings and regional quality layer.	Free/open use but requires proper identification and attribution; direct app calls may expose user IP/geocoordinates to MET.	Identifying User-Agent, avoid unnecessary traffic, attribution, proxy gateway for privacy.
Open-Meteo	MVP forecast API and multi-model access.	Free tier is for non-commercial/evaluation; commercial product should use paid commercial plan or self-host under applicable open-source obligations.	Attribution, API key for customer endpoint, call-budget monitoring.
OpenAIP	Airspace, airports, navigation and aviation-related data where available.	Requires review of current API terms before redistribution or commercial display.	Attribution, freshness timestamp, no official-navigation guarantee.
OpenStreetMap	Base-map data and geographic context.	Use must comply with ODbL/attribution and tile-provider policy; avoid overusing public tile servers.	Visible OSM attribution, own tiles or commercial tile provider at scale.
Copernicus DEM GLO-30/GLO-90	Terrain/slope/aspect/rotor and thermal model features.	Free access for GLO-30/GLO-90 with required notices; review EEA-10/10m restrictions.	Source notice, modified-data notice, DOI/reference in documentation.
Weather stations / webcams	Live reality layer, launch validation, CV analysis.	Depends on station owner and camera owner agreements.	Station-owner consent, data-processing terms, camera privacy review, retention limits.
Pilot reports / IGC tracks	Crowd reality, training labels, debrief.	User-generated content governed by platform ToS and privacy settings.	Contributor licence, deletion rights, anonymization controls, moderation.

3.3 Data-source approval workflow

1. Engineer proposes a source in data_source_licences with status pending.
2. Legal/product owner reviews licence, commercial usage, attribution, rate limits, privacy impact and redistribution rights.
3. Security reviews authentication and supply-chain risk.
4. Data engineering implements source-specific ingestion guardrails and attribution metadata.
5. Source is approved only after automated tests verify attribution, provenance and caching rules.
6. Licence changes trigger a compliance review and may disable affected features until approved.

4. Attribution and Provenance Architecture

Attribution cannot be handled as static footer text. The app combines several sources into derived forecasts and AI summaries. Therefore attribution must be generated dynamically from provenance metadata.

4.1 Attribution service

```
GET /legal/attribution?context=site_forecast&site_id={id}&from={t1}&to={t2}

Response:
```

```
{
  "context": "site_forecast",
  "sources": [
    {"name":"ECMWF", "licence":"CC BY 4.0", "modified":true},
    {"name":"MET Norway", "licence":"CC BY 4.0", "modified":true},
    {"name":"Copernicus DEM GLO-30", "modified":true}
  ],
  "display_text": "Forecast derived from ECMWF, MET Norway and Copernicus DEM data. Modified by Paragliding Intelligence Platform. Sources do not endorse this service."
}
```


4.2 Provenance fields

Entity	Required provenance fields
weather_model_runs	source_key, model_name, run_time, forecast_step, source_url/object_key, ingest_checksum, licence_id
site_forecasts	source_run_ids, processing_version, terrain_version, correction_model_version, generated_at
flyability_scores	site_forecast_id, scoring_policy_version, pilot_profile_version, hard_blockers, confidence_inputs
ai_briefings	retrieved_context_ids, prompt_template_version, model_id, refusal_reason, safety_policy_version
map_tiles	source_layer_ids, tile_generation_version, attribution_bundle_id
airspace_zones	source_key, source_updated_at, imported_at, effective_from, effective_to, confidence_status

5. Safety and Liability Model

The product must not imply guaranteed flight safety. The core legal architecture is a safety boundary that defines what the product is allowed to say, what it must never say, and how uncertainty is displayed.

5.1 Safety boundary language

Allowed language	Forbidden language	Reason
Recommended: GO / MAYBE / NO-GO based on available forecast and site rules.	Safe to fly.	Absolute safety guarantee.
Best forecast window: 12:00-15:00, confidence 78%.	You can launch at 12:00.	Implies permission/control.
Pilot remains responsible for final decision and local observation.	The app authorizes this flight.	The app is not an authority.
Data unavailable / stale; do not rely on this forecast.	Assume good conditions if no data.	Unsafe optimism.
Airspace layer is informational; check official sources.	Airspace cleared.	Official clearance implication.
Emergency sharing assists others but does not replace emergency services.	Rescue guaranteed.	Unacceptable reliance.

5.2 Hard safety blockers

```
if thunderstorm_risk >= policy.thunderstorm_red_threshold:
    return NO_GO("Thunderstorm risk exceeds safety threshold")
if precipitation_intensity >= policy.rain_red_threshold:
    return NO_GO("Rain risk is incompatible with safe paragliding")
if forecast_age_minutes > policy.max_forecast_age_minutes:
    return NO_DATA("Forecast is stale")
if model_disagreement >= policy.max_disagreement_for_go:
    return MAYBE("Forecast models disagree; verify locally")
if launch_wind_direction not in site.allowed_directions:
    return NO_GO("Wind direction incompatible with launch")
if gust_spread >= policy.max_gust_spread:
```

```
    return NO_GO("Gust spread exceeds risk limit")
```

5.3 UI safety disclaimer pattern

Every decision surface uses a short disclaimer near the recommendation, with a full disclaimer accessible through a details link. The disclaimer must be short enough to be seen, not hidden in terms of service.

```
Decision card copy:
"Decision support only. Conditions can change quickly. Verify at launch, check local rules and airspace, and use your own judgement before flying."
```

5.4 Incident evidence retention

Evidence item	Retention proposal	Access control
AI briefing context snapshot	24 months for paid users; shorter for free unless incident flagged.	User, support with approval, legal hold.
Flyability score inputs	24 months.	Internal audit/support.
Weather model provenance	36 months or as storage permits.	Engineering and audit.
Live track history	User-configurable; default limited retention.	User, emergency share recipients, support only on legal basis.
Alert delivery receipts	12-24 months.	Support and incident review.
Terms acceptance version	Account lifetime + statutory period.	Compliance and billing support.

6. Privacy, GDPR and Location Governance

The platform processes location history, live tracking, pilot skill profiles, flight logs, emergency contacts, payment metadata and community reports. The privacy architecture must be designed before launch, not patched later.

6.1 Data classification

Data category	Examples	Risk	Default handling
Public data	Launch names, public site notes, general forecast layers.	Low.	Cache and display with attribution.
Account data	Email, display name, subscription tier.	Medium.	Encrypt in transit, access controlled.
Precise location	Live GPS track, launch/landing positions, route history.	High.	Explicit opt-in, retention controls, sharing controls.
Pilot profile	Skill level, wing class, SIV status, risk tolerance.	Medium/high.	Used for safety personalization; editable and exportable.
Emergency contacts	Names, phone/email, relationship.	High.	Explicit purpose, encrypted storage, limited access.
Payment metadata	Stripe/customer IDs, invoice status.	Medium.	Do not store card data; rely on payment processor.
AI interaction	Briefing questions, retrieved context, generated response.	Medium/high if it contains location.	Minimize, redact where possible, retention by tier/policy.

6.2 Consent events

```
CREATE TABLE consent_events (
  id UUID PRIMARY KEY,
  user_id UUID NOT NULL REFERENCES users(id),
  consent_type TEXT NOT NULL, -- live_tracking, ai_briefing, training_data, emergency_share, marketing
  status TEXT NOT NULL, -- granted, revoked
  policy_version TEXT NOT NULL,
  context JSONB NOT NULL DEFAULT '{}',
  ip_hash TEXT,
  user_agent_hash TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

6.3 Privacy controls by feature

Feature	Privacy requirement	Implementation requirement
Weather map	No account required for basic map where possible.	Anonymous session, coarse analytics, no precise tracking without permission.
Nearby flyable sites	Location permission must be optional.	Allow manual location or city search.
Live tracking	Explicit opt-in for each tracking session.	Prominent stop button, auto-expire share links, session log.
Rescue mode	Explicit emergency-sharing consent where possible.	Limited recipients, audit trail, emergency override policy reviewed by counsel.
IGC upload / debrief	User controls visibility and training-data use.	Private by default; anonymized training toggle separate from debrief.
Community reports	Prevent accidental doxxing of current location.	Report location rounding option and delayed public display option.
Club/school accounts	Separate personal pilot data from organization admin visibility.	RBAC + row-level filters by membership and consent.

6.4 Data subject rights workflow

7. Export: generate machine-readable export of account, pilot profile, tracks, reports, billing metadata references and AI briefings.
8. Delete: delete or anonymize personal data according to legal/accounting retention rules.
9. Correction: allow user to edit profile, emergency contacts, skill level and public display name.
10. Restriction: pause processing for training data or community visibility while preserving account.
11. Portability: export tracks as GPX/IGC where legally and technically possible.
12. Support review: identity verification before sensitive export/delete operations.

7. Terms of Service and User Policy Architecture

Terms of Service should be versioned like software. The platform must store which version each user accepted and which terms were active at the time of a decision, purchase, incident or dispute.

7.1 Terms-version tables

```
CREATE TABLE legal_documents (
  id UUID PRIMARY KEY,
  doc_type TEXT NOT NULL, -- tos, privacy_policy, safety_disclaimer, refund_policy,
data_processing_addendum
  version TEXT NOT NULL,
  locale TEXT NOT NULL DEFAULT 'en',
  effective_at TIMESTAMPTZ NOT NULL,
  content_hash TEXT NOT NULL,
  url TEXT NOT NULL,
  status TEXT NOT NULL DEFAULT 'active'
);

CREATE TABLE user_legal_acceptances (
  id UUID PRIMARY KEY,
  user_id UUID NOT NULL REFERENCES users(id),
  legal_document_id UUID NOT NULL REFERENCES legal_documents(id),
  accepted_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  acceptance_context JSONB NOT NULL DEFAULT '{}'
);
```

7.2 Terms sections required

Section	Purpose
Decision-support disclaimer	Clarifies that recommendations are informational, not permission to fly.
Weather uncertainty	Explains forecasts can be wrong, stale, unavailable or locally inaccurate.

Pilot responsibility	States pilots must verify conditions, skills, equipment, local regulations and airspace.
AI limitations	Explains AI summaries are derived from source data and can omit or misunderstand context.
Emergency limitations	States rescue sharing is best-effort and not a substitute for emergency services.
User-generated content	Defines licence to reports/photos/tracks and moderation rights.
Prohibited use	No commercial aviation reliance, no emergency-service replacement, no unlawful tracking.
Subscription terms	Renewal, cancellation, refunds, trial conditions, app-store handling.
Data rights/privacy	References privacy policy, data export/delete and location controls.
Jurisdiction and dispute resolution	To be decided by legal counsel based on company location and markets.

8. AI Output Policy and Hallucination Liability Controls

AI is useful because it can translate complex weather into pilot-friendly briefings. It is also a liability risk because narrative text can sound more certain than the underlying forecast. The AI layer must be constrained by policy and verified with tests.

8.1 AI grounding contract

```
POST /ai/briefing
{
  "site_id": "...",
  "time_window": {"from":"2026-06-05T10:00Z", "to":"2026-06-05T18:00Z"},
  "pilot_profile_id": "...",
  "required_context": ["site_forecast", "site_rules", "airspace", "alerts", "model_disagreement"]
}

AI service must reject if required_context is unavailable, stale, or below confidence threshold.
No briefing may invent: wind, gust, cloudbase, storm risk, airspace status, launch rules or pilot skill.
```

8.2 AI safe-response policy

Condition	AI response rule
Missing forecast	Do not answer with GO. Explain missing data and recommend checking live sources.
High model disagreement	Return MAYBE/uncertain and explain disagreement.
User asks for guarantee	Refuse guarantee; explain decision-support boundary.
User asks to ignore safety blocker	Refuse and restate blocker.
Airspace unknown	Do not claim airspace is clear.
Medical/rescue emergency	Tell user to contact local emergency services; app data may assist but not replace response.
Beginner profile with marginal conditions	Downgrade to NO-GO or training-only with instructor.

8.3 AI audit log

```
CREATE TABLE ai_safety_audit_logs (
  id UUID PRIMARY KEY,
  user_id UUID,
  briefing_id UUID,
  model_id TEXT NOT NULL,
  prompt_template_version TEXT NOT NULL,
  retrieved_context_hash TEXT NOT NULL,
  output_hash TEXT NOT NULL,
  safety_policy_version TEXT NOT NULL,
  refusal_reason TEXT,
  hard_blockers JSONB NOT NULL DEFAULT '[]',
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

9. Airspace, Maps and Terrain Legal Controls

Airspace and map data are operationally sensitive because stale or incorrect data can create legal and physical risk. The app should display airspace as an informational overlay and preserve source freshness, not as an official navigation clearance system.

9.1 Airspace rules

- Every airspace polygon must display source, imported timestamp and effective date if available.
- Airspace warnings must use cautious language such as "restricted/controlled airspace may be nearby" instead of "clear".
- The app must support region-specific official links for national aviation authorities where available.
- Expired or stale airspace data must not be hidden; it must be flagged and downgrade recommendations.
- Competitions and schools should be able to attach local rules, NOTAM links and site-specific restrictions.

9.2 Map tile compliance

Provider/data	Compliance requirement	Architecture decision
OpenStreetMap database	Visible attribution and ODbL compliance for database use.	Use own tile pipeline or commercial tile provider at scale; do not overload public tiles.
MapLibre renderer	Respect library open-source licence and notices.	Maintain OSS notice file in app and web footer.
Commercial map provider	Follow provider attribution and caching restrictions.	Provider-specific attribution renderer and cache policy.
Terrain DEM	Source and modified-data notices.	Attribution service injects DEM notices on terrain-derived layers.

10. Subscription, Billing and Entitlement Legal Controls

The product uses free and paid tiers. Paid features must be governed by a technical entitlement service and clear customer terms. Avoid hidden safety implications where paid users get more optimistic decisions; paid users may get more data, not weaker safety rules.

10.1 Entitlement safety principle

Paid tiers can unlock more forecast models, longer forecast range, AI briefings, advanced alerts, club dashboards and route planning. Paid tiers must never bypass hard safety blockers. A PRO+ pilot may see more detail, but NO-GO blockers remain universal.

10.2 Billing records

```
CREATE TABLE subscription_entitlements (
  user_id UUID NOT NULL,
  plan_key TEXT NOT NULL, -- free, pro, pro_plus, club, school, event, rescue, api
  entitlement_key TEXT NOT NULL,
  source TEXT NOT NULL, -- stripe, apple, google, admin, enterprise_contract
  status TEXT NOT NULL, -- active, trialing, past_due, cancelled, expired
  valid_from TIMESTAMPTZ NOT NULL,
  valid_until TIMESTAMPTZ,
  evidence_ref TEXT,
  PRIMARY KEY (user_id, entitlement_key)
);
```

10.3 Subscription compliance checklist

13. Show price, billing period, renewal terms and cancellation method before purchase.
14. Use Stripe/app-store customer portal for cancellation where applicable.
15. Do not store raw card data.
16. Implement receipts, invoices and entitlement audit logs.
17. Handle grace periods and failed payment without corrupting safety-critical data access.
18. Provide enterprise contracts for clubs/schools/events/rescue with DPA and service terms.
19. Localize consumer terms for major launch markets before paid expansion.

11. Community Content and Moderation

Crowdsourced reports are a competitive advantage, but they create legal, safety and privacy risks. The platform should treat community observations as unverified unless submitted by trusted users or validated by station/webcam/model data.

11.1 Report trust levels

Trust level	Meaning	UI label
raw	Unverified user report.	Pilot report - unverified
trusted_user	Submitted by a user with verified history or club role.	Trusted local report
sensor_confirmed	Consistent with nearby weather station/webcam/model.	Sensor-confirmed
conflicting	Conflicts with station/model or other reports.	Conflicting reports
moderated	Reviewed by club/site moderator.	Moderator reviewed

11.2 User-generated content licence

Terms should grant the platform enough rights to display, process, translate, moderate and derive safety analytics from reports while preserving user privacy choices. Training-data use should be separate from basic display where possible, especially for location-rich tracks and photos.

12. Emergency and Rescue Mode Policy

Rescue features must be helpful without creating a false promise. The product can provide last known position, recent track, forecast wind, possible drift and shareable emergency links, but it must not claim to replace official emergency services.

12.1 Rescue-mode control contract

Control	Requirement
Activation	User, emergency contact or organization admin can activate only under defined permissions.
Emergency copy	Display: "Call local emergency services first. This tool only shares supporting information."
Link expiry	Emergency share links expire by default and can be revoked.
Audit	All access to rescue data is logged.
Data minimization	Share only relevant track/time window and emergency contact details.
Authority accounts	Rescue organization access requires contract and role-based permissions.

13. Data Retention Schedule

Data type	Default retention	Reason	User control
Anonymous map analytics	30-90 days.	Product analytics.	Opt out where required.
Account profile	Account lifetime.	Service operation.	Edit/delete account.
Precise live tracks	User-selected: 30 days, 1 year, forever; default limited.	Debrief/rescue/history.	Delete/export.
AI briefings	90 days free, 24 months paid; legal hold if incident.	Support, audit, quality.	Delete where legally allowed.
Weather provenance	24-36 months.	Forecast validation and audit.	Not usually personal.

Payment invoices	Statutory accounting period.	Tax/accounting.	Cannot delete before legal period.
Consent/legal acceptance	Account lifetime + limitation period.	Compliance evidence.	Stored as legal evidence.
Community reports	Until deleted or superseded; anonymize old reports.	Site knowledge.	Delete/anonymize request.

14. Contract Pack Required Before Launch

Document	Audience	Owner
Terms of Service	All users.	Legal + Product.
Privacy Policy	All users.	Legal + DPO/Security.
Safety Disclaimer	All users; displayed in app.	Legal + Safety Lead.
Data Processing Addendum	Clubs, schools, enterprise, rescue.	Legal.
Source Attribution Page	All users; regulators/partners.	Data Engineering + Legal.
Refund/Cancellation Policy	Paid users.	Billing/Product.
Acceptable Use Policy	All users/API customers.	Security + Legal.
API Terms	API customers.	Legal + Platform.
Station Owner Agreement	Weather station/webcam partners.	Partnerships + Legal.
Incident Review Policy	Internal/support/legal.	Safety + Security.

15. Legal and Safety Acceptance Gates

Gate	Must pass before
Licence register approved for all production sources.	Public launch.
Attribution renderer verified on web, mobile, map, API and PDF exports.	Public launch.
Safety disclaimer appears on decision surfaces and onboarding.	Beta launch.
AI no-hallucination evaluation passes with zero invented critical weather values.	AI beta.
Hard blockers cannot be bypassed by paid tier, AI or user preference.	Any launch.
Privacy export/delete workflows tested.	Public launch in EU/EEA.
Live tracking consent, stop button and share-link expiry tested.	Tracking launch.
Incident evidence retention policy implemented.	Paid launch.
Subscription cancellation and entitlement reconciliation tested.	Paid launch.
Legal counsel review completed for target countries.	Commercial launch.

16. Risk Register

Risk	Impact	Mitigation
Data source licence changes	Feature shutdown or legal exposure.	Licence monitor, source abstraction, fallback sources.
User treats GO as guarantee	Injury/liability.	Language policy, uncertainty display, disclaimers, conservative scoring.
AI invents condition	Unsafe decision.	RAG-only context, blockers, evaluations, refusal rules.
Stale airspace data	Legal violation by pilot.	Timestamp, official-source links, stale-data warnings.
Live tracking privacy breach	Severe privacy harm.	RBAC, encryption, revocable links, audit logs.
Public tile provider abuse	Service ban.	Own tile infrastructure or paid provider.
Weather API overuse	Blocked data access.	Cache, rate-limit, batch calls, source-specific policies.
Rescue feature overpromises	Reliance risk.	Emergency disclaimer, best-effort policy, official emergency-first flow.
Commercial use without proper API plan	Contract/licence breach.	Entitlement/source register locks commercial features to approved sources.

17. Implementation Backlog

ID	Backlog item	Priority
LEGAL-001	Implement data_source_licences registry and approval workflow.	P0
LEGAL-002	Build attribution service and UI attribution component.	P0
LEGAL-003	Create terms/version acceptance tables and	P0

	API.	
LEGAL-004	Add safety-disclaimer component to all decision surfaces.	P0
LEGAL-005	Implement AI grounding/refusal audit logs.	P0
LEGAL-006	Build consent events and privacy controls for live tracking.	P0
LEGAL-007	Build export/delete/anonymize workflows.	P1
LEGAL-008	Implement emergency-share expiry and access logs.	P1
LEGAL-009	Create source licence monitoring task and review dashboard.	P1
LEGAL-010	Create legal acceptance test suite for release gates.	P1
LEGAL-011	Add OSS and third-party notices to web/mobile.	P1
LEGAL-012	Create station-owner and community-contributor data agreements.	P2

18. Reference Register

The legal/compliance assumptions in this architecture should be checked during implementation against primary sources. The following sources were used as technical and legal design references:

- ECMWF Open Data and ECMWF Terms of Use - open data governed by CC BY 4.0 and source-specific terms.
- MET Norway Weather API Terms of Service - identification, traffic rules, attribution and privacy notes for direct API calls.
- Open-Meteo pricing/licence pages - commercial use licence and API key requirements for commercial/dedicated endpoint usage.
- OpenAIP legal/API materials - airspace and aviation data terms to be reviewed before redistribution.
- OpenStreetMap copyright/licence pages - ODbL and attribution obligations for OSM data use.
- Copernicus DEM documentation - free GLO-30/GLO-90 access and required source notices.
- European Commission GDPR/data-protection guidance - EU personal data protection principles and cross-border safeguards.
- European Commission AI Act pages - risk-based AI framework, timeline and high-risk-system compliance posture.
- Stripe, Apple and Google subscription/payment documentation - billing, receipts and app-store policy integration.

19. Summary

This volume turns legal and safety concerns into implementation requirements. For a paragliding meteo platform to be better than general weather apps, it must be more trustworthy, more transparent and more conservative in uncertainty. The legal architecture is therefore not just paperwork: it directly shapes the scoring engine, attribution service, AI guardrails, privacy controls, live tracking, subscription system and incident review process.

The next recommended volume is Volume 13 - Launch-Site Playbooks and Local Expert Operations. That document should define how each launch site becomes a structured safety knowledge base with local rules, rotor zones, moderator roles, trusted reports and site-specific operational playbooks.